Affordable Detection of Microplastics in Water Using Optical Light Scattering

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Introduction

- Microplastics: plastic fragments <5 mm commonly found in water bodies
- Associated health risks: inflammation, oxidative stress, cellular damage
- 80–94% of drinking water worldwide contains microplastics
- Current techniques (FTIR, Raman, Microscopy) are expensive and lab-dependent
- Need for a low-cost, portable, real-time detection system

Problem Statement

- Microplastics are invisible to the naked eye → difficult to detect quickly
- High-end instruments require skilled operators
- No affordable portable device for preliminary water screening
- Need a simple, reliable system usable in field conditions

Aim: Develop an Arduino-based low-cost optical sensor

Theory & Background

Optical scattering: particles scatter light based on size & shape

Clean water → minimal scattering → high light intensity

Microplastics → more scattering → reduced light intensity

Methodology

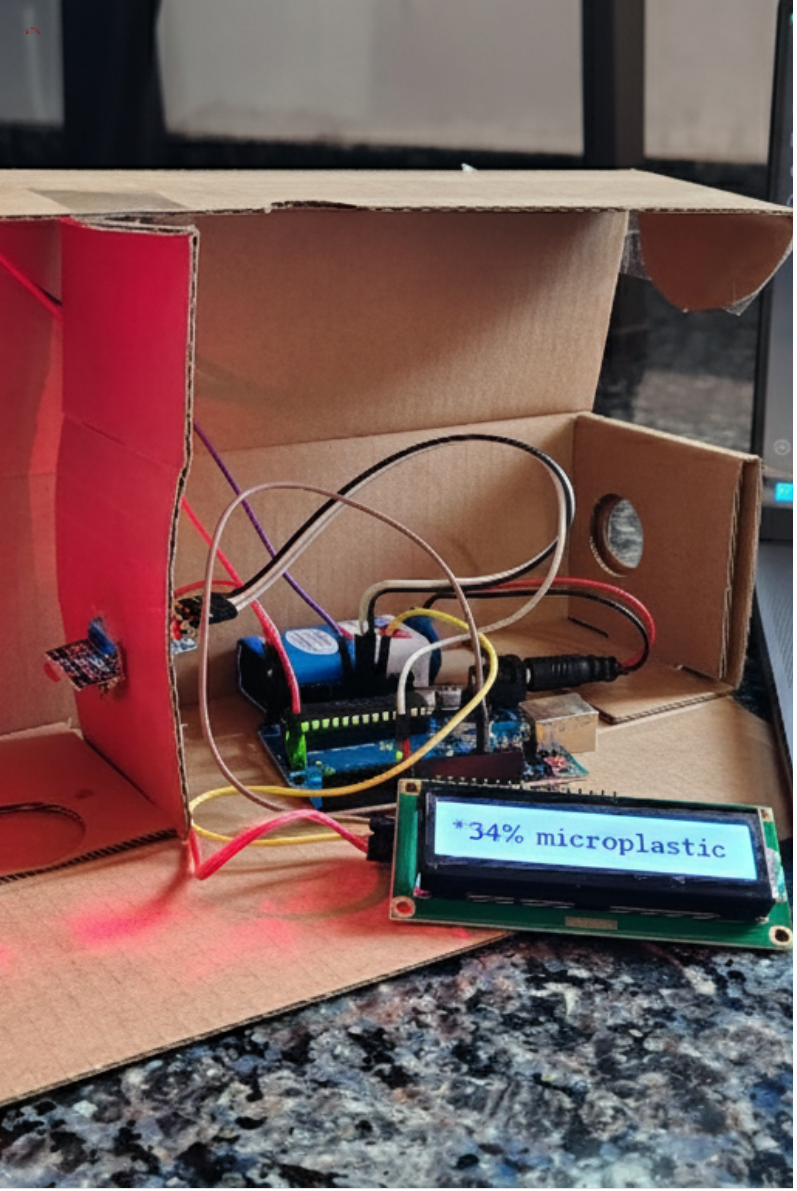
Laser diode emits light through water sample

Photodiode measures transmitted intensity

Arduino processes signal and sends data to LCD

Transparent beaker holds sample

Difference in intensity → clean vs contaminated water



Experimental Setup

Components Used

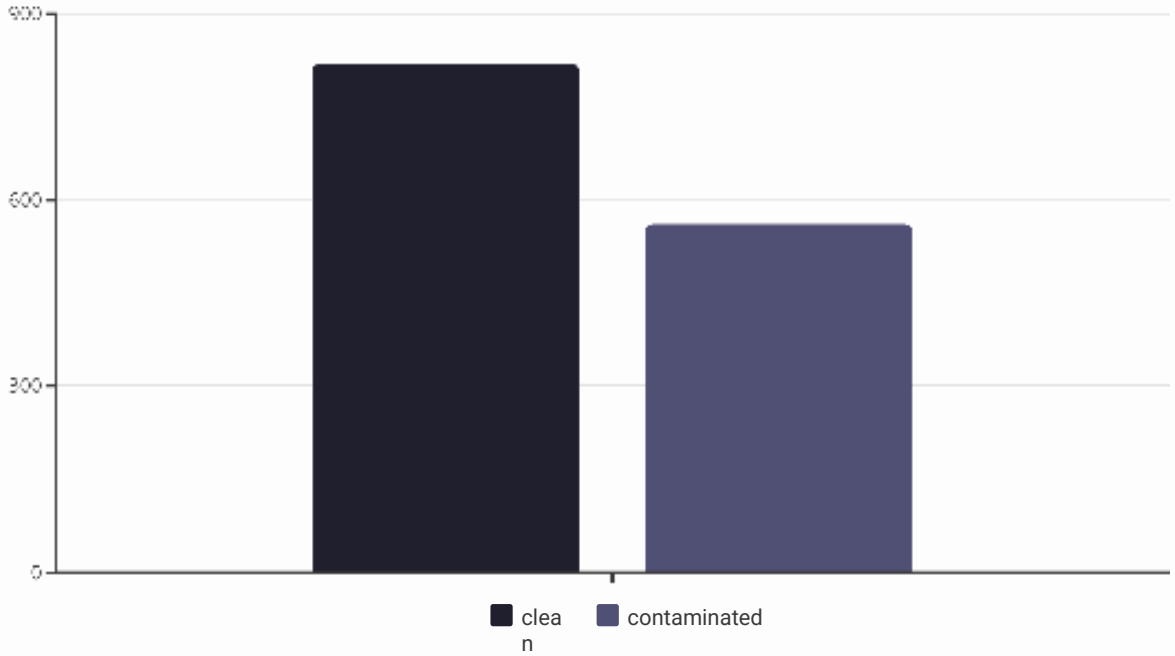
- Laser diode
- Photodiode
- Arduino Uno R3
- LCD module
- Transparent beaker
- Real-time intensity display

Key Features

- Simple prototype assembled using low-cost components
- Total cost: <₹1300 INR
- Portable and compact design
- Easy to use (no expert required)

Results: Light Intensity Comparison

Experimental results of light intensity measurement in ADC units:



820

Clean Water

Higher intensity (ADC units)

560

Contaminated Water

Lower intensity (ADC units)

31.7%

Intensity Reduction

Due to microplastics

Results Table

Parameter	Clean Water	Contaminated Water
Avg Intensity (ADC)	820	560
Intensity Reduction	–	31.7%
Detection Result	Clean	Contaminated
Response Time	2.1s	2.3 s
Repeatability (5 trials)	High	High

- Consistent detection across multiple trials
- Real-time response within ~2 seconds

Discussion

System Performance

- Detects microplastics through measurable reduction in transmitted intensity
- Accuracy ~85% in preliminary tests
- Works well for visible and moderate particle concentration
- Low-cost components make it suitable for rapid screening

Key Limitation

- Not polymer-specific

Advantages, Limitations & Conclusion

Advantages

- Affordable (<₹1300 INR)
- Portable and compact
- Easy to use (no expert required)
- Real-time results

Limitations

- Cannot identify polymer type
- Limited precision for very fine or low-concentration microplastics
- Affected by color/turbidity of water

Conclusion

- Proposed a low-cost, optical-scattering-based microplastic detection system
- Offers fast, real-time results with good accuracy (85%)
- Suitable for preliminary water screening
- Promising alternative to expensive laboratory techniques
- Future scope: AI-driven classification, improved sensitivity, handheld device

References

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THANK YOU

Questions?